Def. Let X, Y be Metric Spaces, and $f_n : X \to Y$ be function defined for each $n \in \mathbb{N}$.

For each $x \in X$, define $f(x) \overset{\text{def}}{=} \lim_{n\to\infty} f_n(x)$ if the limit exists.

This f is called limit function of $\{f_n\}$.

Def. Suppose that a sequence of functions $\{f_n\}$ on X into Y satisfies that;
the limit function f is defined on X and

For all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ s.t. $n \geq N \Rightarrow d_Y(f_n(x), f(x)) \leq \varepsilon$, $\forall x \in E$.

Then, we say that $\{f_n\}$ Converges uniformly on X to f. Denote $f_n \rightrightarrows f$.

Note: $\forall x \in X$, $d_Y(f_n(x), f(x)) \leq \varepsilon \iff \sup_{x\in X} d_Y(f_n(x), f(x)) \leq \varepsilon$. That is,

$$f_n \rightrightarrows f \text{ on } X \iff \lim_{n\to\infty} \sup_{x\in X} d_Y(f_n(x), f(x)) = 0.$$

Thm. Let $\{f_n\}$ be functions defined on any Metric Space X into Complete Metric Space Y. Then,

In General Metric Space

$\{f_n\}$ Converges uniformly on X to f if and only if $\forall \varepsilon > 0$, $\exists N \in \mathbb{N}$ s.t. $m, n \geq N \Rightarrow \sup_{x\in X} d_Y(f_n(x), f_m(x)) \leq \varepsilon$

In Complete Metric space

Proof. If $f_n \rightrightarrows f$, then,

$$\sup_{x\in X} d_Y(f_n(x), f_m(x)) \leq \sup_{x\in X} d_Y(f_n(x), f(x)) + \sup_{x\in X} d_Y(f(x), f_m(x)) \to 0$$

give the result. (간단해 보이지만, Supremum 의 성질과 triangle inequality 가 합쳐진 콤비네이션.)

Conversely, Since Cauchy sequence in the Complete Metric Space Converges, the limit function f exist.
Given $\varepsilon > 0$, choose $N \in \mathbb{N}$ such that the Cauchy condition holds.
For fixed $n \geq N$, for all $x \in X$ and $m \geq N$, $d_Y(f_n(x), f_m(x)) \leq \varepsilon$.
So, letting $m \to \infty$, we obtain $d_Y(f_n(x), f(x)) \leq \varepsilon$, for all $x \in X$. Thus proved. ■

Fail Try.

$$\lim_{m\to\infty} \sup_{x\in X} d_Y(f_m(x), f_n(x)) = \sup_{x\in X} d_Y(f(x), f_n(x))$$

gives the result. (More specifically, let $n \in \mathbb{N}$ fixed. Observe that

$$\left| \sup_{x\in X} d_Y(f_m(x), f_n(x)) - \sup_{x\in X} d_Y(f(x), f_n(x)) \right|$$

$$\sup_{x\in X} d_Y(f(x), f_n(x)) = \sup_{x\in X} d_Y(\lim_{m\to\infty} f_m(x), f_n(x)) \underset{d_Y \text{ Continuous}}{=} \sup_{x\in X} \lim_{m\to\infty} d_Y(f_m(x), f_n(x)) \overset{(*)}{=} \lim_{m\to\infty} \sup_{x\in X} d_Y(f_m(x), f_n(x))$$

$$\left| \sup_{x\in X} \lim_{m\to\infty} d_Y(f_m(x), f_n(x)) - \sup_{x\in X} d_Y(f_m(x), f_n(x)) \right| \leq \sup_{x\in X} \left| \lim_{m\to\infty} d_Y(f_m(x), f_n(x)) - d_Y(f_m(x), f_n(x)) \right| \to 0$$

gives $(*)$.

「Uniform Convergence and Continuity.

「Thm. Suppose that $f_n, f : X \to Y$ are functions where X is Metric, Y is complete Metric sp.
Let $x \in X$ be a limit point of X.
If for each $n \in \mathbb{N}$, the limit $\lim_{t \to x} f_n(t)$ exists and $f_n \to f$ uniformly on X,
then
$$\lim_{t \to x} \lim_{n \to \infty} f_n(t) = \lim_{n \to \infty} \lim_{t \to x} f_n(t)$$

Proof: For each $k \in \mathbb{N}$, take $A_k = \lim_{t \to x} f_k(t)$. And, Since the uniformly convergence, given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$m, n \geq N \Rightarrow d_Y(f_n(t), f_m(t)) \leq \varepsilon \text{ for all } t \in X.$$

And, for such $m, n \geq N$, let $x_0 \in X$ such that

$$d_Y(A_m, f_m(x_0)) < \varepsilon \quad \text{and} \quad d_Y(A_n, f_n(x_0)) < \varepsilon$$

(for some δ_i, $\delta > 0$, $t \in B_\delta(x) \Rightarrow d_Y(f_k(t), A_k) < \varepsilon$, by def. of A_k)
Now,

$$d_Y(A_m, A_n) \leq d_Y(A_m, f_m(x_0)) + d_Y(f_m(x_0), f_n(x_0)) + d_Y(f_n(x_0), A_n) \leq 3\varepsilon,$$

Therefore $\{A_k\}$ is a Cauchy sequence in Y, So it is Convergent.
Let $A = \lim_{n \to \infty} A_n$ Then,

$$d_Y(f(t), A) \leq \underbrace{d_Y(f(t), f_n(t))}_{f_n \to f \text{ uniformly}} + \underbrace{d_Y(f_n(t), A_n)}_{\substack{f_n(t) \to A_n \\ \text{as } t \to x}} + \underbrace{d_Y(A_n, A)}_{A_n \to A.} \to 0$$

Rigorously, given $\varepsilon > 0$, let $N_1, N_2 \in \mathbb{N}$ s.t.

$$n \geq N_1 \Rightarrow d_Y(A_n, A) < \varepsilon \quad \text{and} \quad n \geq N_2 \Rightarrow \forall t \in X,\ d_Y(f_n(t), f(t)) < \varepsilon.$$

Now, take $n \geq \max\{N_1, N_2\}$ and $\delta > 0$ s.t.

$$t \in B_\delta(x) \Rightarrow d_Y(f_n(t), A_n) < \varepsilon. \quad \blacksquare$$

「Corollary. If $\{f_n : X \to Y\}$ is a sequence of Continuous maps on a Metric space X into Complete Y, and $f_n \rightrightarrows f$, then f is Continuous.

「Uniform Convergence with Continuity

「Thm. Suppose that f_n $(n \in \mathbb{N})$ are functions on a Metric X into Complete Metric Y, Let $x \in X$ be a limit point of X.
If for each $n \in \mathbb{N}$, the limit $\lim_{t \to x} f_n(t)$ exists and f_n Converges uniformly on X, then

$$\lim_{t \to x} \lim_{n \to \infty} f_n(t) = \lim_{n \to \infty} \lim_{t \to x} f_n(t).$$

Proof. By the Uniformly Convergence, given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ s.t.

$$n, m \geq N \Rightarrow d_Y(f_m(t), f_n(t)) \leq \varepsilon \text{ for any } t \in X.$$

「Thm. Let K be a Compact space, and $f_n : K \to \mathbb{R}$ be continuous functions which converges pointwise to a continuous function f on K.
If $f_n \geq f_{n+1}$ for all $n \in \mathbb{N}$, then $f_n \to f$ uniformly.

Proof. Put $g_n = f_n - f$. Then, g_n is continuous and $g_n \to 0$ pointwise, and $g_n \geq g_{n+1}$.
Given $\varepsilon > 0$, let $K_n = \{x \in K \mid g_n(x) \geq \varepsilon\} = g_n^{-1}[[\varepsilon, \infty)]$
Since g_n is continuous, K_n is a closed set in a compact space, so it is compact.
And, Since $g_n \geq g_{n+1}$, $K_n \supseteq K_{n+1}$ (if $g_{n+1}(x) \geq \varepsilon$, then $g_n(x) \geq g_{n+1}(x) \geq \varepsilon$)
Fix $x \in X$. Since $g_n(x) \to 0$ as $n \to \infty$, for large enough $n \in \mathbb{N}$, $x \notin K_n$.
That is, $x \notin \bigcap K_n$, so $\bigcap K_n$ is empty. But, Since each K_n is compact, thus $K_N = \emptyset$ for some $N \in \mathbb{N}$. That is,

$$0 \leq g_N(x) < \varepsilon \text{ for all } x \in K,$$

and since $g_N \geq g_{N+1} \geq g_{N+1} \geq \cdots$, so $\sup |g_n - 0| \to 0$. ▨

「Thm. Let X be a Metric Space, and $\mathscr{C}(X)$ be the all Complex-Valued Conti, bounded functions on X.
Define $\|f\| = \sup_{x \in X} \|f(x)\|$. then it is norm on $\mathscr{C}(X)$. This norm can derive a Metric on $\mathscr{C}(X)$,

Uniform Convergence and Integration.

Thm. Let $\alpha : [a,b] \to \mathbb{R}$ be monotone increasing and $f_n \in \mathcal{R}(\alpha)$, $n \in \mathbb{N}$.
If $f_n \rightrightarrows f$ on $[a,b]$, then $f \in \mathcal{R}(\alpha)$ and

$$\lim_{n\to\infty} \int_a^b f_n \, d\alpha = \int_a^b \lim_{n\to\infty} f_n \, d\alpha = \int_a^b f \, d\alpha.$$

Proof. For each $n \in \mathbb{N}$, define $\varepsilon_n = \sup_{x\in[a,b]} |f_n(x) - f(x)|$. Then, clearly

$$f_n - \varepsilon_n \le f \le f_n + \varepsilon_n \qquad (\text{for all } t \in [a,b])$$

Therefore, $$\int_a^b f_n - \varepsilon_n \, d\alpha \le \underline{\int_a^b} f \, d\alpha \le \overline{\int_a^b} f \, d\alpha \le \int_a^b f_n + \varepsilon_n \, d\alpha \qquad (*)$$

Hence $$0 \le \overline{\int_a^b} f \, d\alpha - \underline{\int_a^b} f \, d\alpha \le 2\,\varepsilon_n \cdot [\alpha(b) - \alpha(a)] \to 0 \text{ as } n \to \infty.$$

Thus $f \in \mathcal{R}(\alpha)$. The equality is obtained by $(*)$, using $f \in \mathcal{R}(\alpha)$.

Uniform Convergence and Differentiation.

Thm. Let $f_n : [a,b] \to \mathbb{R}$ be differentiable maps s.t for some $x_0 \in [a,b]$, $\{f(x_0)\}$ converges.
If $\{f_n'\}$ converges uniformly on $[a,b]$, then $\{f_n\}$ converges uniformly on $[a,b]$ to f with

$$\frac{d}{dx} \lim_{n\to\infty} f_n(x) = \lim_{n\to\infty} \frac{d}{dx} f_n(x).$$

Proof. Given $\varepsilon > 0$, take $N \in \mathbb{N}$ s.t $n, m \ge N$ implies

$$|f_n(x_0) - f_m(x_0)| \le \frac{\varepsilon}{2} \text{ and } |f_n'(t) - f_m'(t)| \le \frac{\varepsilon}{2(b-a)}, \ \forall t \in [a,b].$$

By the Mean-Value Theorem for $f_n - f_m$, for all $x, t \in [a,b]$,

$$\left| f_n(x) - f_m(x) - \big(f_n(t) - f_m(t)\big) \right| \le \frac{\varepsilon}{2(b-a)} \cdot (b-a) = \frac{\varepsilon}{2}. \qquad (*)$$

Therefore, for any $x, t \in [a,b]$ and $n, m \ge N$,

$$|f_n(x) - f_m(x)| \le |f_n(x) - f_m(x) - f_n(x_0) + f_m(x_0)| + |f_n(x_0) - f_m(x_0)| \le \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

so that $\{f_n\}$ converges uniformly, let $f = \lim_{n\to\infty} f_n$. For each $x \in [a,b]$, define on $[a,b] \setminus \{x\}$

$$\phi_n(t) = \frac{f_n(t) - f_n(x)}{t - x} \quad \text{and} \quad \phi(t) = \frac{f(t) - f(x)}{t - x}$$

$(*)$ gives ϕ_n is Cauchy sequence, thus converges uniformly, to ϕ. Finally, the fact that ϕ_n continuous gives the result.

Summary. Let $f_n : X \to Y$ be functions, $f_n \to f$.

Continuity: If f_n are continuous and $f_n \rightrightarrows f$, then f is continuous.

Differentiation: If f_n are differentiable and $f_n' \rightrightarrows f'$ and $f_n(x_0) \to f(x_0)$, then $f_n \rightrightarrows f$ and

$$\frac{d}{dx} \lim_{n\to\infty} f_n(x) = \lim_{n\to\infty} \frac{d}{dx} f_n(x)$$

Integral: If $f_n \in \mathscr{R}(\alpha)$ and $f_n \rightrightarrows f$, then $f \in \mathscr{R}(\alpha)$ and

$$\int_a^b \lim_{n\to\infty} f_n \, d\alpha = \lim_{n\to\infty} \int_a^b f_n \, d\alpha$$

「Counter examples

「 $f_n : [0,1] \to \mathbb{R} ; x \mapsto x^n$

f_n conti,

$$\lim_{n\to\infty} f_n(x) = \begin{cases} 0 & \text{if } 0 \le x < 1 \\ 1 & \text{if } x = 1. \end{cases}$$

f not conti

「 $f_n : [0,1] \to \mathbb{R} ; x \mapsto \frac{1}{n} x$

$\bar{f}_n : \mathbb{R} \to \mathbb{R} ; x \mapsto \frac{1}{n} x$

f_n converges uniformly

$\bar{f}_n$ not converges uniformly

「 $f_n : [0,2] \to \mathbb{R} ; x \mapsto \dfrac{1-x}{1+x^n}$

$$\lim_{n\to\infty} f_n(x) = \begin{cases} 1-x & \text{if } 0 \le x \le 1 \\ 0 & \text{if } 1 < x \le 2. \end{cases}$$

f_n differentiable, converges uniformly,

but

f not differentiable at $x=1$

In $[0,1]$, $|f_n(x) - f(x)| = \left| (1-x)\cdot\left(\frac{1}{1+x^n} - 1\right) \right| = \dfrac{|1-x|\cdot|x^n|}{|1+x^n|} \le |1-x|\cdot|x^n| = (1-x)\cdot x^n.$

Since $((1-x)\cdot x^n)' = -x^n + (1-x)n\cdot x^{n-1} = nx^{n-1} - (n+1)x^n = x^{n-1}(n-(n+1)x)$

has a maximum at $x = \frac{n}{n+1}$, $\left(1 - \frac{n}{n+1}\right)\cdot\left(\frac{n}{n+1}\right)^n = \dfrac{n^n}{(n+1)^{n+1}} \to 0$ as $n \to \infty$.

So, $\sup\limits_{[0,1]} |f_n(x) - f(x)| \to 0$ as $n \to \infty$.

In $[1,2]$, $|f_n(x) - f(x)| = \left| \dfrac{1-x}{1+x^n} \right| = \dfrac{x-1}{1+x^n} \le \dfrac{x-1}{x^n} = \dfrac{1}{x^{n-1}} - \dfrac{1}{x^n} \to 0$ as $n \to \infty$.

Now, $f_n \to f$ converges uniformly.

「 $f_n : \mathbb{R} \to \mathbb{R} ; x \mapsto \frac{1}{\sqrt{n}} \cdot \sin(nx).$

$f \equiv 0$, since $|\sin(nx)| \le 1$

$f_n' : \mathbb{R} \to \mathbb{R} ; x \mapsto \sqrt{n} \cdot \cos(nx)$

f_n differentiable, converges uniformly,

but

f_n' not converges pointwise.

⌈ Let $X_n = \{x_n \mid n \in \mathbb{N}\} = \mathbb{Q} \cap [0,1]$.

$$f_n : [0,1] \to \mathbb{R} : x \mapsto \begin{cases} 1 & \text{if } x \in X_n \\ 0 & \text{if } x \notin X_n. \end{cases}$$

f_n Riemann-integralable, since it has finite disconti

but

$$f : [0,1] \to \mathbb{R} : x \mapsto \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q} \end{cases}$$

f not integrable, it is Dirichlet.

⌈

$$f_n : [0,1] \to \mathbb{R} : x \mapsto \begin{cases} n & \text{if } 0 < x < \frac{1}{n} \\ 0 & \text{if } x = 0 \text{ or } \frac{1}{n} \le x \le 1. \end{cases}$$

$\lim\limits_{n\to\infty} f_n \equiv 0$, but $\int_0^1 f_n = 1$. Cannot change limit under the integral.

⌈

$$f_n : \mathbb{R} \to \mathbb{R} : x \mapsto \frac{x^2}{(1+x^2)^n}$$

f_n continuous, but the sum is not continuous.

$$f(x) = \sum_{n=0}^{\infty} f_n(x) = \sum_{n=0}^{\infty} \frac{x^2}{(1+x^2)^n} = \begin{cases} 0 & \text{if } x = 0 \\ 1+x^2 & \text{if } x \neq 0 \end{cases} \qquad \left(x^2 \cdot \frac{1}{1 - \frac{1}{1+x^2}} \right)$$

⌈ $f_n : [0,1] \to \mathbb{R} : x \mapsto nx \cdot (1-x^2)^n$ f_n differentiable,

$\lim\limits_{n\to\infty} f_n(x) = 0$

$$\int_0^1 x(1-x^2)^n \, dx = \int_1^0 -\frac{1}{2} \cdot t^n \, dt = \frac{1}{2} \cdot \frac{1}{n+1} t^{n+1} \Big|_0^1 = \frac{1}{2} \cdot \frac{1}{n+1}.$$

So, $\int_0^1 f_n(x)\, dx = \frac{1}{2} \cdot \frac{n}{n+1} \to \frac{1}{2}$, but $\int_0^1 f \, dx = 0$,

Note; replace nx to n^2x, then $\int_0^1 f_n \to \infty$.